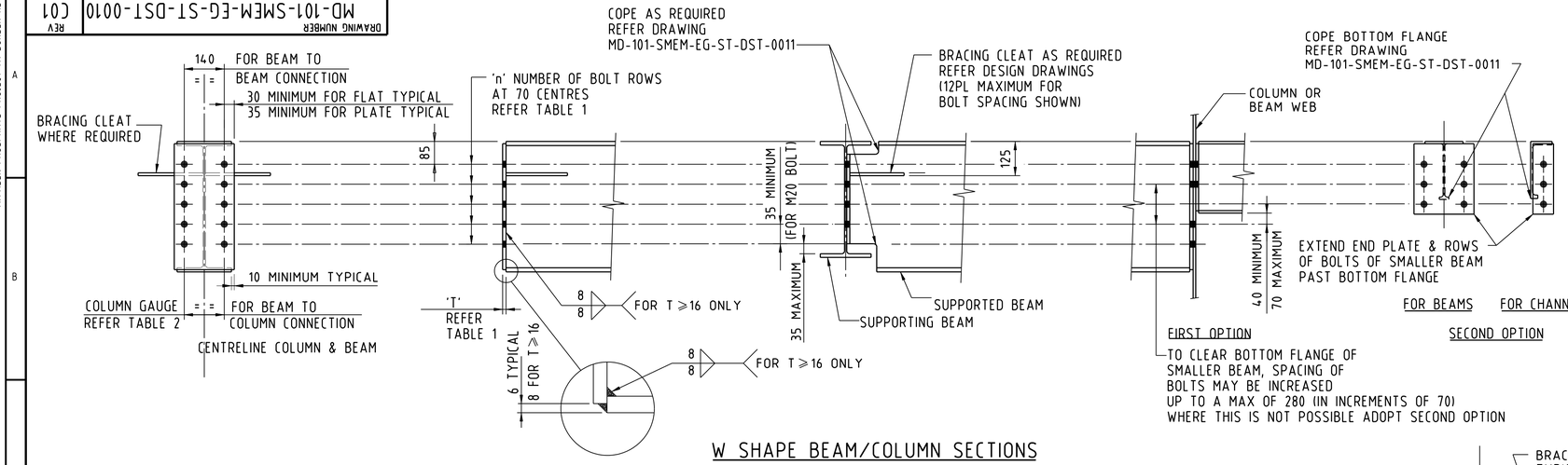
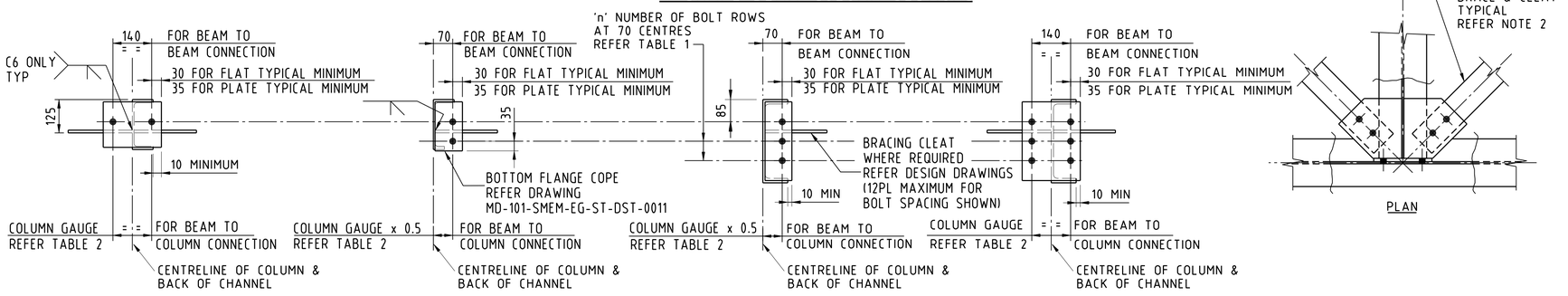
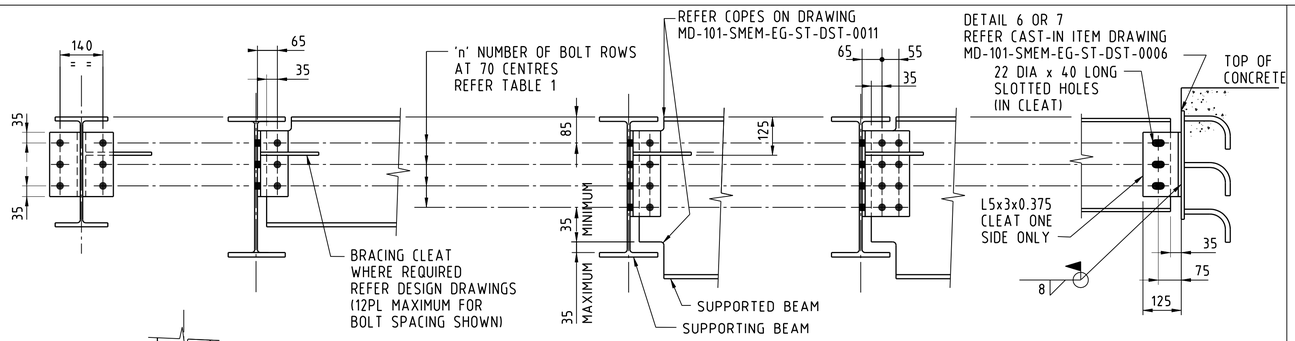


W SHAPE BEAM/COLUMN SECTIONS

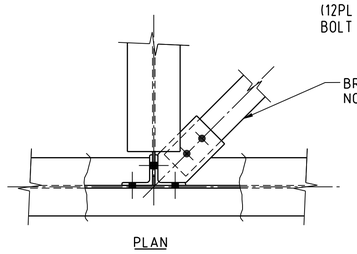


CHANNELS C6&C7 | OPTION 'A' FOR CHANNEL C6 | OPTION 'A' FOR CHANNELS C7-C15 | CHANNELS C8-C15 | TYPICAL FLOOR BRACING CONNECTION

FULL HEIGHT END PLATE DETAILS (FEP)



TYPE ACA | TYPE ACB & ACC | TYPE ACD



TYPICAL HORIZONTAL BRACING CONNECTION

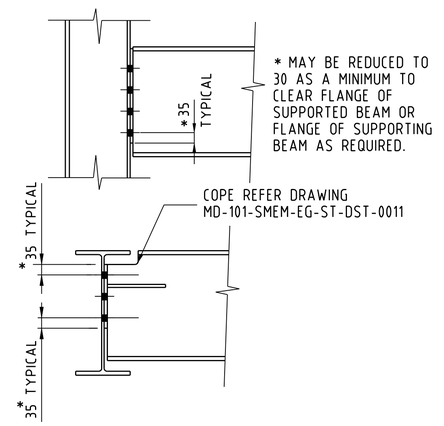
ANGLE CLEAT DETAILS

TABLE 3		
TYPE	ANGLE SIZE	BOLT DIA
ACA	L4x4x0.3125	20
ACB	L6x3.5x0.375	20
ACC	L6x4x0.625	24
ACD	L5x3x0.375	20

LEGEND

AC/n/p

- DENOTES BOLTING PROCEDURE
- DENOTES NUMBER OF BOLT ROWS
- DENOTES DOUBLE ANGLE CLEAT (TO BE ACA UNO REFER TABLE 3)



NOTE: DETAILS AS PER FULL HEIGHT END PLATES (FEP) EXCEPT AS NOTED

PARTIAL END PLATE DETAILS (PEP)

TABLE 1				
BEAM MEMBER SIZE	NUMBER OF BOLT ROWS 'n' UNO: FULL HEIGHT END PLATE OR ANGLE CLEAT	FULL HEIGHT END PLATE OPTION 'A'	WEB SIDE PLATE	END PLATE THICKNESS 'T' UNO
C6x13	1	2	2	10
C7x12.25	1	2	2	10
C8x18.75	2	2	2	10
C9x20	2	2	2	10
C10x25	2	2	2	10
C12x20.7	3	3	3	10
C15x33.9	3	3	3	10
W6x20	1	-	2	10
W8x35	2 ✳	-	2	10
W10x49	2	-	2	10
W12x72	3	-	3	10
W12x106	3	-	3	10
W14x159	3	-	3	10
W14x193	3	-	3	10
W8x18	2 ✳	-	2	10
W10x22	2	-	2	10
W10x26	2	-	2	10
W12x26	3	-	3	10
W12x30	3	-	3	10
W14x34	3	-	3	10
W16x40	4	-	4	10
W18x46	4	-	5	10
W21x50	5	-	6	10
W21x62	5	-	6	10
W24x68	6	-	7	10
W24x84	6	-	7	10
W27x94	6	-	8	10
W30x116	7	-	9	10
W33x118	8	-	10	12
W36x150	9	-	11	16
W36x170	9	-	11	16
W40x183	10	-	12	16

TABLE 2		
COLUMN MEMBER SIZE	COLUMN GAUGE: ON COLUMN FLANGE	ON COLUMN WEB
C6x13	90	-
C7x12.25	90	-
C8x18.75	90	70
C9x20	90	70
C10x25	90	90
C12x20.7	90	90
C15x33.9	90	90
W6x20	90	50
W8x35	140	90
W10x49	140	90
W12x72	140	90
W12x106	140	90
W14x159	140	90
W14x193	140	90
W8x18	70	90
W10x22	70	90
W10x26	70	90
W12x26	90	90
W12x30	90	90
W14x34	90	90
W16x40	90	90
W18x46	90	90
W21x50	90	90
W21x62	140	90
W24x68	140	90
W24x84	140	90
W27x94	140	140
W30x116	140	140
W33x118	140	140
W36x150	140	140
W36x170	140	140
W40x183	140	140

✳ NOTE WHEN BEAM BOTTOM FLANGE COPING IS REQUIRED
NUMBER OF BOLT ROWS SHALL BE 1.

LEGEND (TO BE USED FOR DETAIL DESIGNATION ON DESIGN DRAWINGS FOR NON STANDARD CONNECTIONS)

PEP/A/n/p/CF/T

- T FOLLOWED BY A NUMBER DENOTES END PLATE THICKNESS
- DENOTES COMPENSATING FLANGE (IF REQUIRED). REFER COPES DRAWING MD-101-SMEM-EG-ST-DST-0011
- DENOTES BOLTING PROCEDURE (OTHER THAN S)
- DENOTES NUMBER OF BOLT ROWS OTHER THAN STANDARD
- DENOTES OPTION WHERE APPLICABLE
- DENOTES PARTIAL HEIGHT END PLATE (FEP DENOTES FULL HEIGHT END PLATE WHERE SHOWN ON DESIGN DRG.)

NOTES:

1. FOR STEELWORK GENERAL NOTES REFER DRAWING MD-101-SMEM-EG-ST-0009
2. FOR TYPICAL BRACING CLEAT AND BRACE DETAILS REFER DRAWING MD-101-SMEM-EG-ST-0012 & 0013
3. STRUCTURAL CAPACITIES OF THE CONNECTIONS SHOWN ON THIS DRAWING SHALL BE CHECKED BY THE CONTRACTOR TO DETERMINE THEIR SUITABILITY FOR THE INTENDED USE.
4. TABLE 1, 2 AND 3 REFER TO MD-101-SMEM-EG-ST-SPC-0001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD

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MA'ADEN معادن
Saudi Arabian Mining Company

DRAWN	J BADAL	07.03.11
CHECKED	J EDWARDS	18.04.12
ENGINEER	D KIRKWOOD	18.04.12
APPROVED	R COVELLA	18.04.12
MA'ADEN		
SCALE NTS		A1

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
BEAM AND BRACING CONNECTIONS SHEET 1

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0010 REV C01

MA'ADEN PHOSPHATE PROJECT-A1 BORDER-REV A02

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